TCP round trip time, timeout

— X —

Math

Random Variable

A random variable is a real-valued function defined over a sample space.

(R.V.E.1)

Example:

A school consists of 7 teachers of whom 4 are male and 3 are female. A committee of 2 teachers is to be formed. If Y stands for the Number of male teachers selected, then Y is a random variable assuming the values 0, 1 and 2.

Events	seqence of Events	$Y = y$
E_1	Male, Male	$Y = 2$
E_2	Male, Female	$Y = 1$
E_3	Female, Female	$Y = 0$

Y: A random variable

y: The value of that Random ~~value~~ variable Y.

Types of R.V

(1) continuous R.V. (2) Discrete R.V.

Discrete Random Variable :

A R.V. defined over a discrete sample space (i.e. that may take only finite or countable number of different isolated values) is referred to as a discrete R.V.

Examples :

1. No. of telephone calls recieved in a telephone both.
2. No. of correct answers in an MCQ type exam
3. No. of bubbles in a glass.
4. No. of children in a family

Continuous Random Variable :

A Random variable defined over a continuous sample space (i.e. that may take any value

Con → Integration
Dis → summation

in a ~~constant in~~ certain interval or collection of intervals) is referred to as a continuous R.V.

Examples :

① Time taken to serve a customer in a Bank counter
② Weight of a six month old baby.
③ Rate of interest offered by a bank.
④ Temperature recorded by the Meteorological Dept.

Probability Distribution :

The idea of a probability distribution exactly parallels that of a frequency distribution. Each type of distribution is based on a set of mutually exclusive and collectively exhaustive measurement classes or class intervals.

(exam 3) 27.2.24

Defn of Probability Distribution :

Any statement of a function associating each of of a set of mutually exclusive and collectively exhaustive classes or class intervals with its probability is a probability Distribution.

Discrete Probability Distribution :

A coin is tossed 3 times

$S = \{ \underset{3}{HHH}, \underset{2}{HHT}, \underset{2}{HTH}, \underset{1}{HTT}, \underset{2}{THH}, \underset{1}{THT}, \underset{1}{TTH}, \underset{0}{TTT} \}$

X : No. of heads obtained

Values of $X = x$	0	1	2	3	
$P(X = x)$	$1/8$	$3/8$	$3/8$	$1/8$	

Math

Discrete Probability Distribution: $\longrightarrow$ Probability Mass Function (PMF)

continuous " " $\longrightarrow$ Probability Density " (PDF)

Discrete R.V: Σ - based

Cont^s R.V: $\int$ - based

Def^n: If a Random variable X has a discrete distribution, the probability distribution of X is defined as the function "f" such that, for any real number x, $f(x) = P(X=x)$.

Conditions of being a PMF:

(I) $f(a) \geq 0$

(II) $\sum_x f(x) = 1$

(III) $P(X = x) = f(x)$

Example:

Verify whether the following functions are PMFs or not

(a) $f(x) = \frac{1}{8}(2x-1)$; $x = 0, 1, 2, 3$

(b) $f(x) = \frac{1}{16}(x+1)$; $x = 0, 1, 2, 3$

(c) $f(x) = \frac{1}{21}(3x+6)$; $x = 1, 2$

Sol^n:

(a) Given, $f(x) = \frac{1}{8}(2x-1)$; $x = 0, 1, 2, 3$

$X = x$	0	1	2	3
$P(X=x) = f(x)$	$-1/8$	$1/8$	$3/8$	$5/8$

summing up the function over the entire range of X

$$\sum_{x=0}^{3} f(x) = f(0) + f(1) + f(2) + f(3)$$
$$= -1/8 + 1/8 + 3/8 + 5/8 = 1$$

since, $P(X=0) = f(0) = -1/8 < 0$, it contradicts the condition (i)

Hence, $f(x)$ is not a PMF

b) sol^n:

$X=x$	0	1	2	3
$P(X=x) = f(x)$	$1/16$	$1/8$	$3/16$	$1/4$

$f(x) \geq 0 \quad \forall x$ (for all)

$\sum f(x) = \frac{1}{16} + \frac{1}{8} + \frac{3}{16} + \frac{1}{4} \neq 1$ [violates (ii) condition]

c) sol^n:

$X=x$	1	2
$P(X=x) = f(x)$	$\frac{3}{7}$	$4/7$

(R.V.E.1)

Example:

Obtain the probability function for the school-teacher problem

X : No of male teachers

Total = 7 post = 2

Male = 4

Female = 3

		$X=x$
E_1	Male, Male	2
E_2	Male, female	1
E_3	Female, Female	0

$$f(2) = P(X=2) = \frac{{}^4C_2}{{}^7C_2} = 6/21$$

$$f(1) = P(X=1) = \frac{{}^4C_1 * {}^3C_1}{{}^7C_2} = 12/21$$

$$f(0) = P(X=0) = \frac{{}^3C_2}{{}^7C_2} = 3/21$$

shortcut formula

$$f(x) = \frac{{}^4C_x \cdot {}^3C_{2-x}}{{}^7C_2}$$

$x = 0, 1, 2$

$X=x$	0	1	2
$P(X=x) = f(x)$	$3/21$	$12/21$	$6/21$

This function is a PMF

Problem : A bag contains 10 balls of ~~what~~ which 4 are black. If $\boxed{3 \text{ balls}}$ are drawn at random without replacement, obtain the probability distribution for the number of black balls drawn.

Sol^n :

X : No. of black balls drawn

$x = 0, 1, 2, 3$

$$f(0) = P(X=0) = \frac{{}^4C_0 * {}^6C_3}{{}^{10}C_3} = \frac{20}{120}$$

$$f(1) = P(X=1) = \frac{{}^4C_1 * {}^6C_2}{{}^{10}C_3} = \frac{60}{120}$$

$$f(2) = P(X=2) = \frac{{}^4C_2 * {}^6C_1}{{}^{10}C_3} = \frac{36}{120}$$

$$f(3) = P(X=3) = \frac{{}^4C_3 * {}^6C_0}{{}^{10}C_3} = \frac{4}{120}$$

$X=x$	0	1	2	3
$P(X=x) = f(x)$	$\frac{20}{120}$	$\frac{60}{120}$	$\frac{36}{120}$	$\frac{4}{120}$

will be PMF.

Problem: The discrete R.V Y has a Probability distribution as shown below:

$Y = y$	-3	-2	-1	0	1
$P(Y=y) = f(y)$	0.10	0.25	0.30	0.15	k

find (i) k (ii) $P(-3 < Y < 0)$ (iii) $\cancel{P(Y \geq -1)}$ $P(Y \geq -1)$

Sol^n:

(i) Since $f(y)$ is a probability distribution, it must satisfy the condition $\sum_y f(y) = 1$

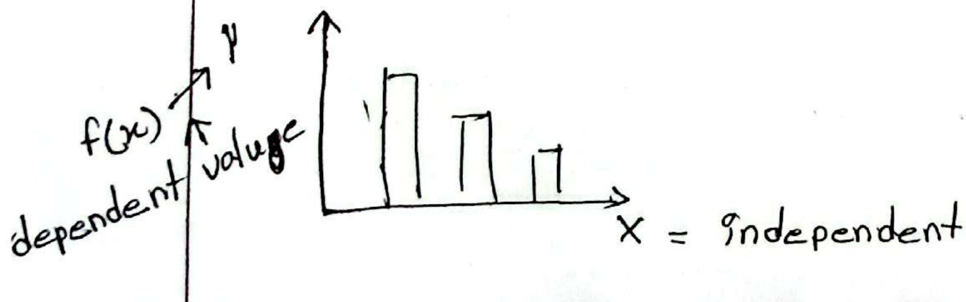
$$\Rightarrow f(-3) + f(-2) + f(-1) + f(0) + f(1) = 1$$

$$\Rightarrow 0.10 + 0.25 + 0.30 + 0.15 + k = 1$$

$$\therefore k = 0.20$$

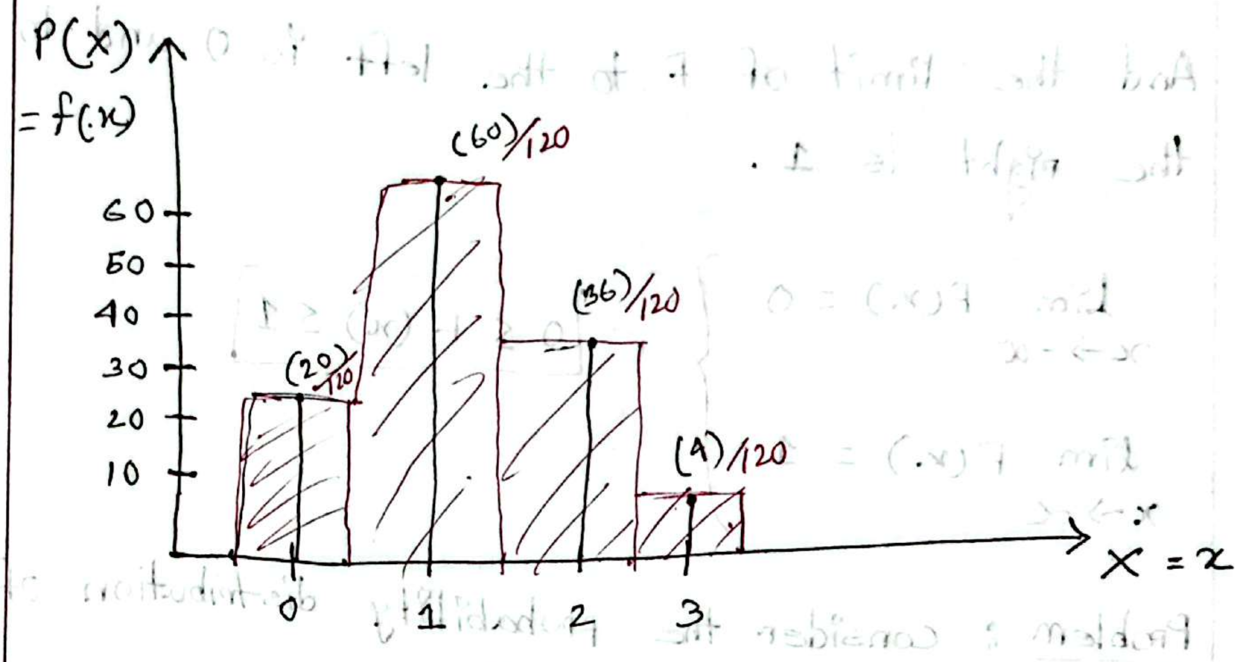
(ii) $P(-3 < Y < 0) = f(-2) + f(-1) = 0.25 + 0.30 = 0.55$

(iii) $P(Y \geq -1) = f(-1) + f(0) + f(1) = 0.30 + 0.15 + 0.20$
$= 0.65$



Graphical Representation of a Discrete Probability Distribution :

$X=x$	0	1	2	3
$P(X=x) = f(x)$	$\frac{20}{120}$	$\frac{60}{120}$	$\frac{36}{120}$	$\frac{4}{120}$



Def^n : Cumulative Distribution Function :

The cumulative distribution function (CDF) or simply the distribution function $F(x)$ of a discrete random variable X with Probability function $P(X=x) = f(x)$ defined over all real numbers x is the cumulative probability upto & including the point x.

Symbolically, we can write

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

The function $F(x)$ is a <u>monotonic increasing</u> function.

i.e. $\boxed{F(a) \leq F(b)}$ for all $a \leq b$

And the limit of F to the left is 0 and to the right is 1.

$$\lim_{x \to -\infty} F(x) = 0$$

$$\lim_{x \to \infty} F(x) = 1$$

$$\boxed{0 \leq F(x) \leq 1}$$

<u>Problem</u> : consider the probability distribution of the Random variable X as follows :

$X = x$	0	1	2	3
$P(X=x) = f(x)$	$20/120$	$60/120$	$36/120$	$4/120$

Here, $F(x_i) = P(X \leq x)$

$F(0) = P(X \leq 0) = f(0) = \frac{20}{120}$

$F(1) = P(X \leq 1) = f(0) + f(1) = \frac{20}{120} + \frac{60}{120} = \frac{80}{120}$

$F(2) = P(X \le 2) = F(0) + F(1) + F(2) = \ldots = 116/120$

$F(3) = \ldots = \frac{120}{120} = 1$

$X = x$	<0	0	1	2	3	>3
$P(X=x) = f(x)$	0	$\frac{20}{120}$	$\frac{80}{120}$	$\frac{36}{120}$	$\frac{4}{120}$	0
$F(x)$	0	$\frac{20}{120}$	$\frac{80}{120}$	$\frac{116}{120}$	$\frac{120}{120} = 1$	1

we can write the values as piecewise function too :

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{20}{120}, & 0 \le x < 1 \\ \frac{80}{120}, & 1 \le x < 2 \\ \frac{116}{120}, & 2 \le x < 3 \\ 1; & x \ge 3 \end{cases}$$

Example :

A continuous R.V X has the function

$$f(x) = \begin{cases} k\,\frac{x^2}{\theta^3} ; & \text{for } 0 \le x \le \theta \\ 0 ; & \end{cases}$$

for some $\theta > 0$

a). For what value of k the function f is a PDF ?

P.M.F

① $f(x) \ge 0$

② $\sum_x f(x) = 1$

③ $\int_{-\infty}^{\infty} f(x)\,dx$

$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\theta} f(x\ ) = dx + \int_{\theta} f(x)\,dx$

$= 0 + \int_0^\theta f(x)\,dx + 0$

$$\int_{x=0}^{\theta} f(x)\,dx = 1 \Rightarrow \int_0^\theta k\,\frac{x^2}{\theta^3}\,dx = 1$$

$$\Rightarrow \frac{k}{\theta^3}\left[\frac{x^3}{3}\right]_0^\theta = 1$$

$$\Rightarrow \frac{k}{\theta^3}\left[\frac{\theta^3}{3}\right] = 1$$

$$\Rightarrow \frac{1}{3}k = 1$$

$\therefore\ k = 3$

$X, Y \longrightarrow$ Discrete

$X = x \, ; \quad Y = y$

$X=x$	0	1	2	3	4
$P(X=x) = f(x)$					

$X=x$ \ $Y=y$	0	1	2	
0	$f(0,0) = P(X=x, Y=y)$	$f(0,1)$	$f(0,2)$	$\Sigma f(0,y)$
1	$f(1,0)$	$f(1,1)$	$f(1,2)$	$\Sigma f(1,y)$
2	$f(2,0)$	$f(2,1)$	$f(2,2)$	$\Sigma f(2,y)$
	$\Sigma f(x,0)$	$\Sigma f(x,1)$	$\Sigma f(x,2)$	

$$P(X=1, Y=2) = f(1,2)$$

Joint Distribution

<u>Def^n:</u> let (X,Y) be a two dimentional discrete random variable with range space R_X and R_Y of X and Y, respectively. Then the function f defined by $f(x,y) = P(X=x, Y=y)$ for all $(x,y) \in R_X \times R_Y$

- from above example:

$$X = \{0,1,2\} \rightarrow R_X \qquad Y = \{0,1,2\} \rightarrow R_Y$$

$$X \times Y = \{0,1,2\} \times \{0,1,2\}$$

$$= \{(0,0), (0,1), (0,2) \ldots \}$$

is called discrete joint probability function or discrete bivariate probability function. If (x, y) is not one of the possible values of (X, Y) then $f(x, y) = 0$

A joint probability function for discrete R.Vs satisfying following three properties:

(I) $f(x, y) \geq 0$ for all $(x, y) \in X \times Y$

(II) $\sum\limits_{\text{All } (x,y)} f(x, y) = 1$

(III) $P(X = x, Y = y) = f(x, y)$ for all (x, y).

Ex: A box contains 4 white balls, 3 black balls and 2 red balls. 2 balls are to be drawn without replacement. Let X denotes the number of white balls and Y denotes the number of black balls drawn. Find the joint probability function of (X, Y). Find the probability that $X + Y \geq 3$. Also find the probability that $X = 1$.

$\underline{\text{sol}^n}$: X = Num white balls $\quad x = 0, 1, 2$

Y = " black balls $\quad y = 0, 1, 2$

$$f(0,0) = P(X=0, Y=0) = \frac{{}^4C_0 \cdot {}^3C_0 \cdot {}^2C_2}{{}^9C_2} = \frac{1}{36}$$

$$f(0,1) = P(X=0, Y=1) = \frac{{}^4C_0 \cdot {}^3C_1 \cdot {}^2C_1}{{}^9C_2} = \frac{1}{6}$$

$$f(0,2) = P(X=0, Y=2) = \frac{{}^4C_0 \cdot {}^3C_2 \cdot {}^2C_0}{{}^9C_2} = \frac{1}{12}$$

$$f(1,0) = P(X=1, Y=0) = \frac{{}^4C_1 \cdot {}^3C_0 \cdot {}^2C_1}{{}^9C_2} = \frac{2}{9}$$

$$f(1,1) = P(X=1, Y=1) = \frac{{}^4C_1 \cdot {}^3C_1 \cdot {}^2C_0}{{}^9C_2} = \frac{1}{3}$$

$$f(1,2) = P(X=1, Y=2) = 0$$

$$f(2,0) = P(X=2, Y=0) = \frac{{}^4C_2 \cdot {}^3C_0 \cdot {}^2C_0}{{}^9C_2} = \frac{1}{6}$$

$$f(2,1) = 0$$

$$f(2,2) = 0$$

Joint Probability function of $f(x,Y)$

$x \downarrow$ \ $y \rightarrow$	0	1	2	
0	$1/36$	$1/6$	$\frac{1}{12}$	Σ
1	$2/9$	$1/3$	0	Σ
2	$1/6$	0	0	Σ

Rowsum $\downarrow$

Colsum $\longrightarrow$ $\longrightarrow$ $\underline{1}$

$$P(X+Y \geq 3) = f(1,2) + f(2,1) + f(2,2)$$
$$= 0$$

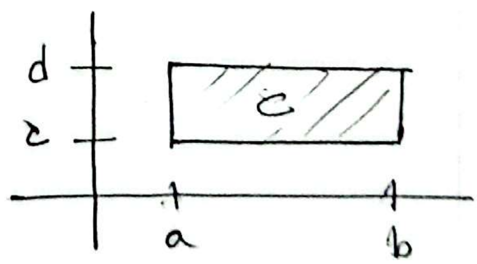
$P(X=1) =$

$\underline{\text{Def}^n}$: If for a continuous bivariate random variables (X,Y), there exists a non-negative function f, defined on the entire xy-plane such that every subset C of pairs of real numbers

$$P[(X,Y) \in C] = \iint_{C} f(x,y)\,dx\,dy$$

$y=c$
$x=a$

$(a,b) \times (c,d) \Leftarrow$



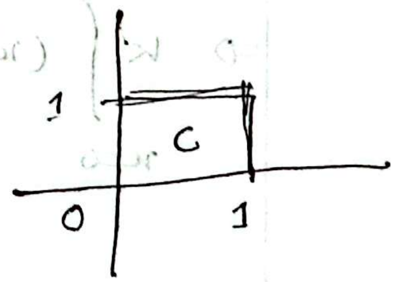
$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x,y)\,dy\,dx = 1 \qquad f(x,y) \geq 0$$

$$P[(X,Y) \in C] = P(a < X < b,\ c < Y < d)$$

$$= \int_{x=a}^{b} \int_{y=c}^{d} f(x,y)\,dy\,dx$$

<u>Ex:</u> Let the joint PDF of the continuous bivariate R.Vs (X, Y) be

$$f(x,y) = \begin{cases} k(x^2 + 2xy); & 0 \leq x \leq 1;\ 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



a) Determine the value of k

b) Find $P(X \leq \frac{1}{2}, Y \geq \frac{1}{2})$

(c) $P(X \leq Y) = ?$

Sol^n:

a) For being a probability density function,

$$\int_{x=-\infty}^{\infty}\int_{y=-\infty}^{\infty} f(x,y)\,dy\,dx = 1$$

$$\Rightarrow \int_{x=0}^{1}\int_{y=0}^{1} k\,(x^2+2xy)\,dy\,dx = 1$$

$$\Rightarrow k\int_{x=0}^{1} \left[x^2y + xy^2\right]_0^1 dx = 1$$

$$\Rightarrow k\int_{x=0}^{1} (x^2+x)\,dx = 1 \quad \Rightarrow k\left[\frac{x^3}{3}+\frac{x^2}{2}\right]_0^1 = 1$$

$$\Rightarrow k\left[\frac{1}{3}+\frac{1}{2}\right] = 1$$

$$\Rightarrow k \cdot \frac{5}{6} = 1$$

b) $P\left(X \leq \frac{1}{2}, Y \geq \frac{1}{2}\right) = \int_{x=0}^{1/2} \int_{y=1/2}^{1} \frac{6}{5}(x^2+2xy)\, dy\, dx = \frac{11}{80}$

(c) $P(X \leq Y) = \int_{y=0}^{1} \int_{x=0}^{y} \frac{6}{5}(x^2 + 2xy)\, dx\, dy = \frac{2}{5}$

Marginal Distribution :

We have noted that if the joint probability function & $f(x,y)$ of two random variables X & Y is known, the probability function of X and Y can be derived separately. In this case, when the distribution of X is derived from $f(x,y)$. the resulting distribution is called the marginal distribution of X.

similarly the probability distribution of Y derived from $f(x,y)$ is known as the marginal distribution of Y

Marginal Distribution of $X = g(x) = \sum_{y} f(x,y)$ ⎫ discrete
" " " $Y = h(y) = \sum_{x} f(x,y)$ ⎭ RVs.

when X & Y are continuous RVs

$$g(x) = \int_{y=\infty}^{\infty} f(x,y)\,dy \quad ; \quad \text{for } -\infty < x < \infty$$

$$h(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx \quad ; \quad \text{for } -\infty < y < \infty$$

Marginal distributions are indeed probability distributions since they satisfy all the properties of a probability distribution.

For continuous case,

$$\int_{-\infty}^{\infty} g(x)\,dx = \int_{x=-\infty}^{\infty}\left[\int_{y=-\infty}^{\infty} f(x,y)\,dy\right]dx = 1$$

and $\mathbb{P}[a<X<b] = \mathbb{P}[a<X<b, -\infty<Y<\infty]$

$$= \int_{x=a}^{b}\int_{y=-\infty}^{\infty} f(x,y)\,dy\,dx = \int_{a}^{b} g(x)\,dx$$

<u>example</u> : suppose X & Y have the following joint probability distribution

x ↓ \ y →	0	1	Row sum
0	0	1/8	1/8
1	1/8	2/8	3/8
2	2/8	1/8	3/8
3	1/8	0	1/8
col sum	4/8	4/8	1

Find the marginal distribution of X & Y

$g(x) = ?$ $h(y) = ?$

Sol^n : For the R.V X

$$g(0) = \mathbb{P}(X=0) = \sum_{y} f(0,y) = f(0,0) + f(0,1) = 1/8$$

$$g(1) = \mathbb{P}(X=1) = \sum_{y} f(1,y) = f(1,0) + f(1,1) = 3/8$$

$$g(2) = \mathbb{P}(X=2) = \sum_{y} f(2,y) = f(2,0) + f(2,1) = 3/8$$

$$g(3) = \mathbb{P}(X=3) = \sum_{y} f(3,y) = f(3,0) + f(3,1) = 1/8$$

Marginal distribution of X is represented by the following table :

$X = x$	0	1	2	3	sum
$P(X=x) = f(x)$	1/8	3/8	3/8	1/8	1

Marginal distribution of Y is represented by the following table :

$Y = y$	0	1	sum
$\mathbb{P}(Y=y) = f(y)$	4/8	4/8	1

$$h(0) = \sum_x f(x,0) = f(0,0) + f(1,0) + f(2,0) + f(3,0)$$

$$= 0 + 1/8 + 2/8 + 1/8 = 4/8$$

$$h(1) = \sum_x f(x,1) = f(0,1) + f(1,1) + f(2,1) + f(3,1) = 1/8 + 2/8 + 1/8 + 0$$

$$= 4/8$$

Example: Find the marginal densities of X & Y from the following joint density function and verify that marginal distributions are also probability distributions

$$f(x,y) = \begin{cases} \frac{1}{8}(6-x-y); & 0<x<2; \quad 2<y<4 \\ 0; & \text{otherwise} \end{cases}$$

also, compute $P(X+Y<3)$ & $P(X<1.5, Y<2.5)$

Soln : By defn the marginal density of X is given by

$$g(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy = \int_{2}^{4} \frac{1}{8}(6-x-y)\,dy = \frac{1}{8}(6-2x)$$

$$= \frac{1}{4}(3-x);$$

$$0 < x < 2$$

$$h(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx = \int_{0}^{2} \frac{1}{8}(6-x-y)\,dx = \frac{1}{4}(5-y)$$

for $2 < y < 4$

$g(x) = \frac{1}{4}(3-x) \geq 0$ for all $x \in (0, 2)$

$h(y) = \frac{1}{4}(5-y) \geq 0$ for all $y \in (2, 4)$

Also, $\int_{x=0}^{2} g(x)\,dx = \int_{0}^{2} \frac{1}{4}(3-x)\,dx = 1$

$$\int_{y=2}^{4} h(y)\,dy = \int_{2}^{4} \frac{1}{4}(5-y)\,dy = 1$$

$2 < y < 4$

$0 < x < 2$

$x + y < 3$

$y < 3 - x$

Now,

$$\mathbb{P}[X+Y<3] = \int_{x=0}^{2}\int_{y=2}^{2-x} \frac{1}{8}(6-x-y)\,dy\,dx =$$

$$\mathbb{P}[X<3/2,\ Y<5/2] = \int_{x=0}^{3/2}\int_{y=2}^{5/2} \frac{1}{8}(6-x-y)\,dy\,dx =$$

Conditional Distribution

The conditional distributions are exactly analogous to the conditional probability of the type. $\mathbb{P}(A|B)$ or $\mathbb{P}(B|A)$, where A and B are two events in the sample space.

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}$$

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B\cap A)}{\mathbb{P}(A)}$$

$$f(y|x) = \mathbb{P}(Y=y \mid X=x) = \frac{\{\mathbb{P}(X=x, Y=y)\}}{\mathbb{P}(X=x)} = \frac{f(x,y)}{g(x)}$$

$$f(x|y) = \mathbb{P}(X=x \mid Y=y) = \frac{\{\mathbb{P}(X=x, Y=y)\}}{\mathbb{P}(Y=y)} = \frac{f(x,y)}{h(y)}$$

$$\boxed{f(y|x) = \frac{f(x,y)}{g(x)}} \quad \& \quad \boxed{f(x|y) = \frac{f(x,y)}{h(y)}}$$

Problem: The following table shows the joint probability distribution of X & Y where X & Y are discrete R.Vs.

$x \downarrow$ \ $y \rightarrow$	0	1	2
0	3/28	6/28	1/28
1	9/28	6/28	0
2	3/28	0	0
	h(0) =	h(1) =	h(2) =

Find $f(x|1)$?

$f(y|1)$?

$\mathbb{P}(X=0 \mid Y=1)$?

$\underline{\text{sol}^n}$: By def^n

$$f(x|1) = \frac{f(x,1)}{h(1)} \quad \text{———— (1)}$$

$$f(x|1) = \frac{f(x,1)}{h(1)} = \left[\frac{7}{3} \times f(x,1)\right] \text{ for } x = 0,1,2$$

$$f(0|1) = \tfrac{7}{3}\, f(0,1) = \frac{7}{3} \times \frac{6}{28} = \frac{1}{2}$$

$$f(1|1) = \frac{7}{3}\, f(1,1) = \frac{7}{3} \times \frac{6}{28} = \frac{1}{2}$$

$$f(2|1) = \frac{7}{3}\, f(2,1) = \frac{7}{3} \times 0 = 0$$

$$h(1) = \sum_x f(x,1) = f(0,1) + f(1,1) + f(2,1)$$

$$= 6/28 + 6/28 + 0 = 3/7$$

$$h(y) = \sum_x f(x,y) =$$

Hence the conditional distribution of X, given Y = 1 is shown in the following table.

x	0	1	2
$f(x\|1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

$$g(1) = \sum_{y} f(1,y) = f(1,0) + f(1,1) + f(1,2)$$

$$= \frac{9}{28} + \frac{6}{28} + 0 = \frac{15}{28}$$

$$f(y|1) = \frac{f(1,y)}{g(1)} = \frac{28}{15} f(1,y). \quad ; \quad \text{for } y = 0, 1, 2$$